

Project Report

1. INTRODUCTION

1.1 Project Overview

This project analyzes the dietary habits and food preferences of college students using Tableau. The dashboard provides interactive visualizations to explore breakfast consumption, eating-out habits, vegetable intake, exercise frequency, and comfort food preferences. The analysis helps identify eating patterns and supports healthier lifestyle decisions through data-driven insights.

1.2 Purpose

The purpose of this project is to analyze college students' food choices and lifestyle behaviors using Tableau dashboards. It aims to convert raw survey data into meaningful visual insights that help understand nutritional habits and support better dietary decision-making.

2. IDEATION PHASE

2.1 Problem Statement

Many college students develop unhealthy eating habits because of busy schedules, academic stress, and lack of nutritional awareness. There is a need to analyze these dietary patterns to understand food preferences and encourage healthier lifestyles.

<https://docs.google.com/document/d/1qHFwBb-mUHucMhIfseyhmp>

[pwHpF0Ht353tEwZR_zZFE/edit?usp=sharing](https://docs.google.com/document/d/1yEvGh3OXmMkAvSwog7lIR905opP6HrQQv7uy2M-qPq8/edit?usp=sharing)

2.2 Empathy Map Canvas

- **Think:** Wants healthy food but has limited time.
- **Feel:** Busy, stressed, and concerned about maintaining good health.
- **Say:** Healthy food is important but difficult to maintain regularly.
- **Do:** Eats outside frequently, skips breakfast, and exercises irregularly.
- **Pain:** Poor diet, unhealthy food choices, and limited nutrition awareness.
- **Gain:** Better understanding of healthy eating habits through data analysis.

<https://docs.google.com/document/d/1yEvGh3OXmMkAvSwog7lIR905opP6HrQQv7uy2M-qPq8/edit?usp=sharing>

2.3 Brainstorming

- Analyze breakfast habits.
- Study eating-out frequency.
- Compare exercise and dietary habits.
- Identify comfort food preferences.
- Create interactive dashboards for better visualization.

<https://docs.google.com/document/d/1yEvGh3OXmMkAvSwog7lIR905opP6HrQQv7uy2M-qPq8/edit?usp=sharing>

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Users upload the dataset, interact with Tableau dashboards, apply filters, view charts, analyze dietary trends, and generate meaningful insights.

3.2 Solution Requirement

Functional Requirements

- Import dataset into Tableau.
- Clean and preprocess data.
- Create dashboards and stories.
- Apply filters and calculated fields.
- Generate interactive visualizations.

Non-Functional Requirements

- Easy to use.
- Fast dashboard loading.
- Accurate visualizations.
- Reliable performance.
- Interactive interface.

[cLca0e3k3ftUThCIOkA/edit?usp=sharing](https://docs.google.com/document/d/11tJPBUJefBnElzAyNKm7iVUx3NDTSG2pxN874KOjKbc/edit?usp=sharing)

3.3 Data Flow Diagram

Dataset → Data Preprocessing → Tableau → Dashboard Creation → Story Creation → Insights & Reports

<https://docs.google.com/document/d/11tJPBUJefBnElzAyNKm7iVUx3NDTSG2pxN874KOjKbc/edit?usp=sharing>

3.4 Technology Stack

<i>Component</i>	<i>Technology</i>
Visualization Tool	Tableau Public
Dataset	CSV
Data Cleaning	Tableau Data Source
Documentation	Google Docs / MS Word
Platform	Windows

https://docs.google.com/document/d/1JpF3iPeDsl4zqdLWBwIVivCv-6ZT_ol9LfFPXgR5tM/edit?usp=sharing

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Tableau dashboard effectively visualizes dietary behaviors and helps identify food consumption trends among college students.

https://docs.google.com/document/d/1sIpV_8i5i1Tvz1VGrYitzWQ0VUp2USosFgA7eOTNRag/edit?usp=sharing

4.2 Proposed Solution

Develop an interactive Tableau dashboard to analyze breakfast habits, eating-out frequency, vegetable consumption, exercise habits, and comfort food preferences.

https://docs.google.com/document/d/1_IWyp66D3rk66N8NyaNvYGZnRQcv4q9abb1GzMqSoSw/edit?usp=sharing

4.3 Solution Architecture

CSV Dataset → Data Cleaning → Tableau → Charts & Dashboards
→ Story → Insights

https://docs.google.com/document/d/1nCSg5SysOm6A90g0zoONGyM4PhDB-sHls_j2nL_tfEQ/edit?usp=sharing

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Sprint 1 – Dataset Collection & Preprocessing
- Sprint 2 – Dashboard Development
- Sprint 3 – Story Creation & Analysis
- Sprint 4 – Testing, Documentation & Deployment

<https://docs.google.com/document/d/1MV0D9qQVw6RLMn40nLTLQVsQu-pCY50kycEXQWeZ--E/edit?usp=sharing>

https://docs.google.com/document/d/162h-yXsluQTneIOI9_WnFurbVsC2M8DF5OjQY-yLnFQ/edit?usp=sharing

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

- Dataset successfully loaded (120 records).
- Data preprocessing completed successfully.
- Filters worked correctly.
- Dashboard rendered without errors.
- Story navigation tested successfully.
- Overall performance: **Successful**.

<https://docs.google.com/document/d/1QTt2JI5TvMt-z4GuTa7NUVA>

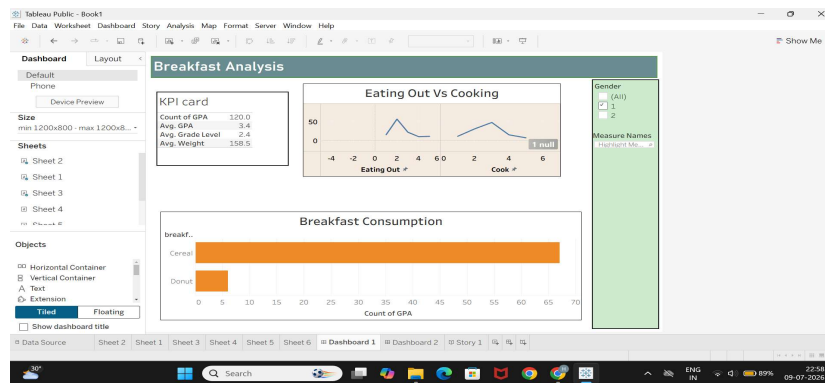
G6pnJlG_0pqF_83o5_S4/edit?usp=sharing

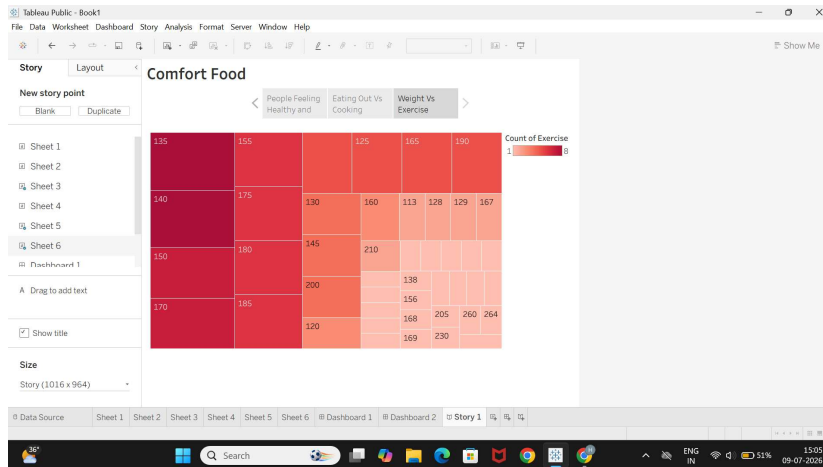
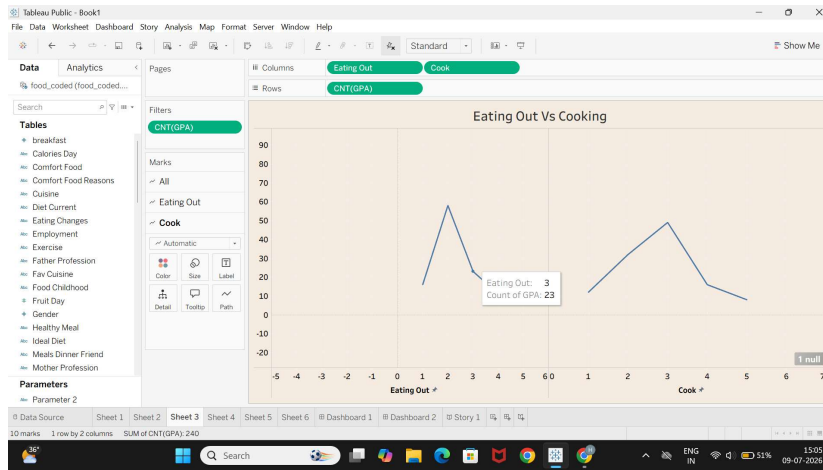
7. RESULTS

7.1 Output Screenshots

Include screenshots of:

- *Dashboards*





8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboards.
- Easy data visualization.
- Quick analysis.
- Better understanding of dietary habits.
- Supports informed decision-making.

Disadvantages

- Limited to the available dataset.
 - Cannot predict future behavior.
 - Requires Tableau knowledge for modifications.
 - Results depend on data quality.
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9. CONCLUSION

The project successfully analyzed college students' food choices using Tableau. Interactive dashboards and stories helped identify dietary patterns, breakfast habits, eating-out behavior, vegetable intake, exercise frequency, and comfort food preferences. The project demonstrates how data visualization can transform survey data into meaningful insights that support healthier lifestyle decisions.

10. FUTURE SCOPE

- Integrate real-time health data.
 - Apply machine learning for prediction.
 - Develop a mobile dashboard.
 - Include larger datasets from multiple colleges.
 - Create personalized diet recommendations.
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11. APPENDIX

Dataset Link

https://drive.google.com/file/d/1ctfCeEQuiLv4FK1aqR_BF1n0t5Mq69PR/view?usp=sharing

GitHub Link

<https://github.com/HarikaD123/College-Food--Choices-Ta-bleau-Project>

Project Demo Link

https://docs.google.com/document/d/152TQ7DuVRxP-1rZi3_JPhCaVOY4x5ygJ15flz-0D_ck/edit?usp=sharing